Cliff of RSA
Forest of public key
River of ECC
Sea of classical cryptography
The Realm of post-quantum
CURVE 25519

Sword of Academia
Sword of Industry
pqc
State of the Art
pqc

Migrating a Silicon Root of Trust to Post-Quantum Crypto

Amin Abdulrahman, Evan Apinis, Andrew 'bunnie' Huang, Matthias J. Kannwischer, Ruben Niederhagen, Felix Oberhansl,
Hoang Nguyen Hien Pham, Jade Philipoom, Dominic Rizzo, Robert Schilling, Peter Schwabe, Tobias Stelzer, Augustine Tang, Andreas Zankl

1. SPHINCS+ Secure Boot

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2. HW/SW Codesign for ML-KEM & ML-DSA

1. SPHINCS+ Secure Boot
2. HW/SW Codesign for ML-KEM & ML-DSA
3. Ongoing Work & Lessons Learned

SPHINCS+

(aka SLH-DSA)

CHES 2022 🇧🇪

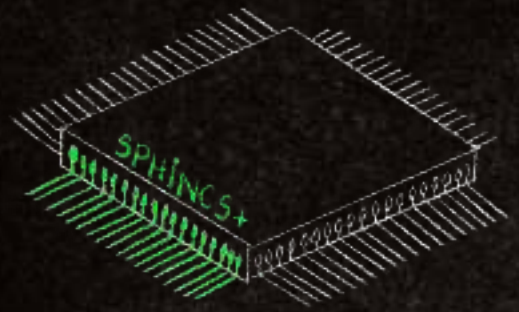
Peter you should take a look at this open-source silicon design !!!

OK!!!

29 hours later...

The first signature is VERIFIED

6 months later... SPHINCS+ secure boot!



3x faster than unmodified reference implementation

- shake-128s-simple: < 13 ms @100MHz
- sha3-128s-simple: ~ 9.3 ms @100MHz

← Blog post: Landing SPHINCS+ on OpenTitan

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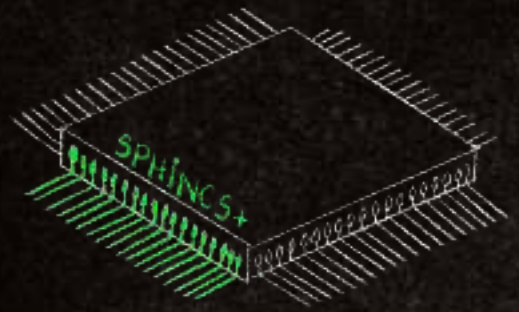

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6x slower than RSA-3072

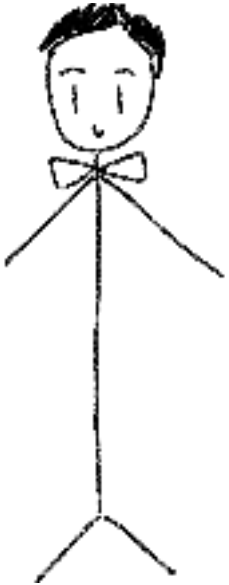
3x slower than ECDSA-P256

- one-time programmable (OTP) memory configuration to enable

← Blog post: Landing SPHINCS+ on OpenTitan

HW/SW Codesign for ML-KEM & ML-DSA

Amin's Master Thesis: Dilithium on OpenTitan (2023)



8 bits	8 bits	8 bits	8 bits	8 bits	8 bits	8 bits	8 bits

Proposed vector instructions for OpenTitan Big-Number Accelerator (OTBN):

Thesis

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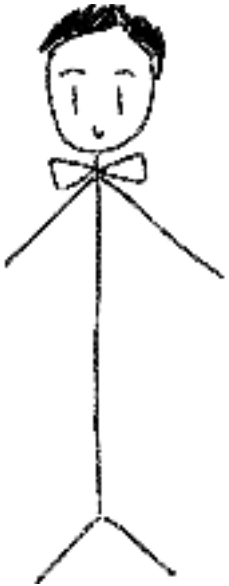

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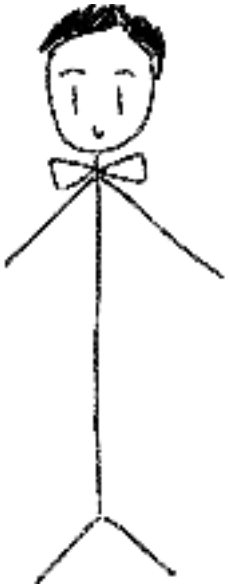

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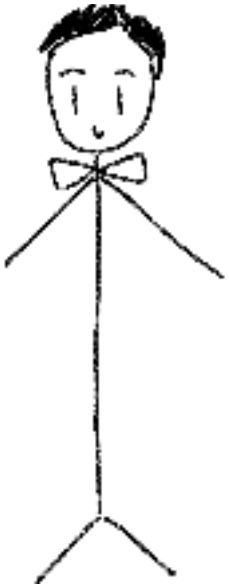

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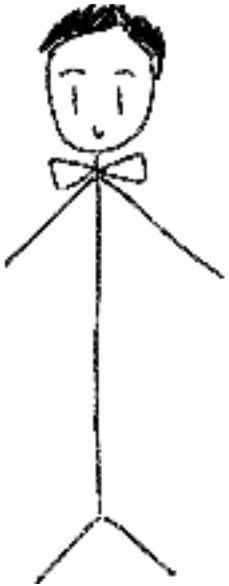

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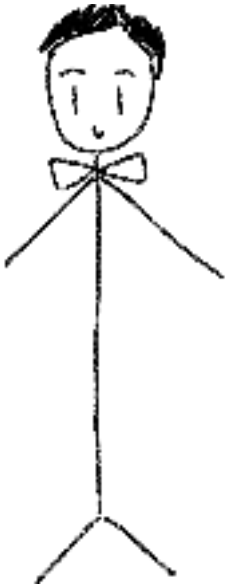

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Idea for SHA3/SHAKE connection to OTBN

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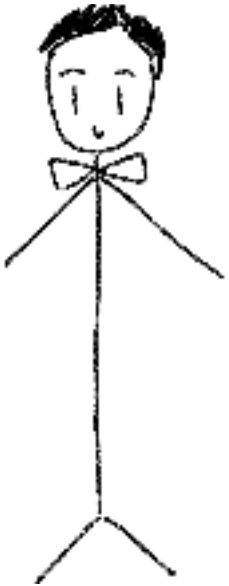

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Thesis

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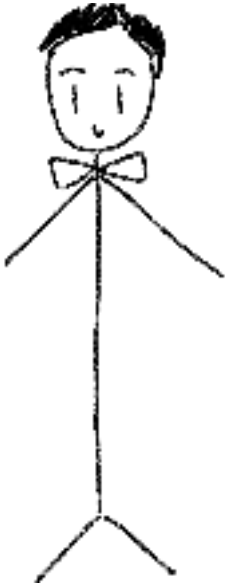

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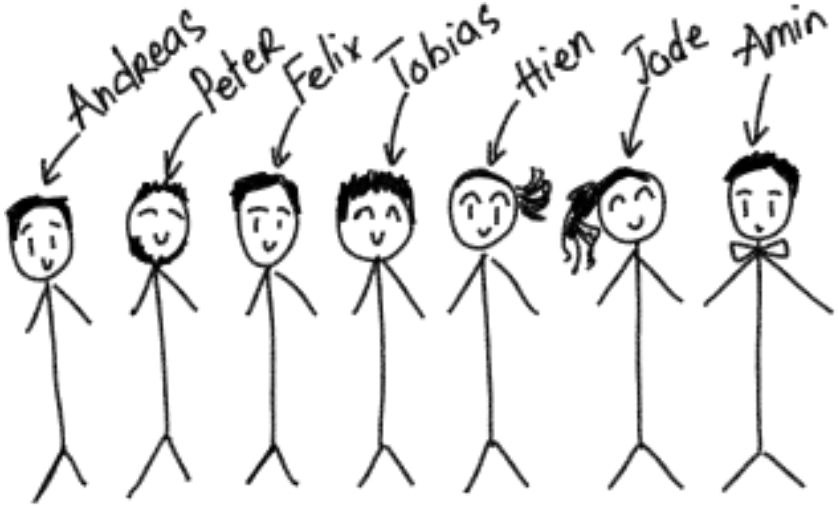
Software implementation of Round-3 Dilithium

Increased IMEM from 8 to 32KB and DMEM from 4 to 128KB

Thesis

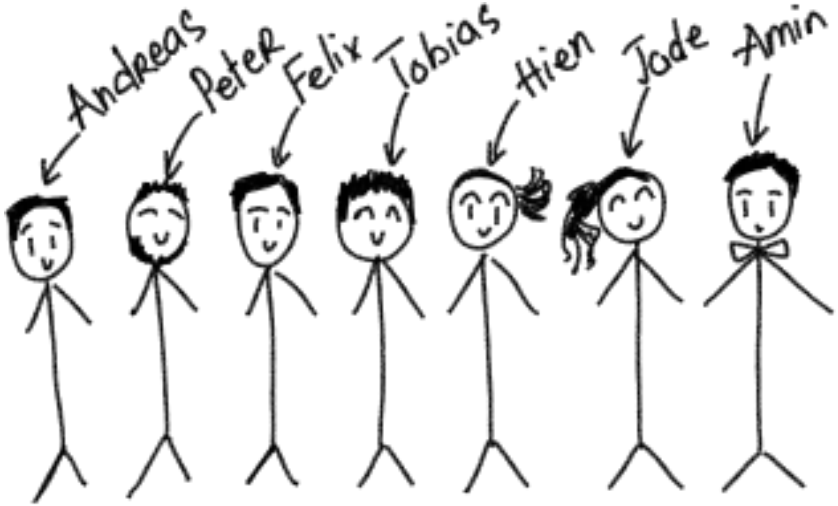
Let's collaborate again



Vector instructions for OTBN:

- bn.{add,sub,mul}v(m)
- bn.shv

Paper Code



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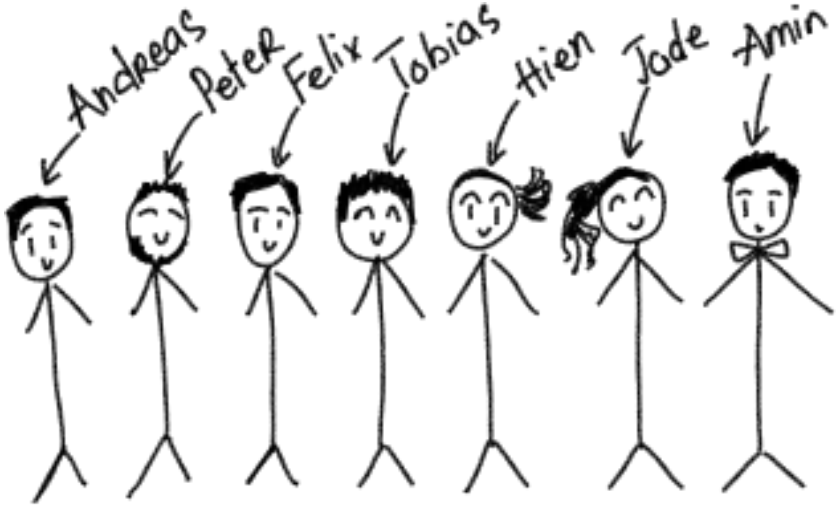

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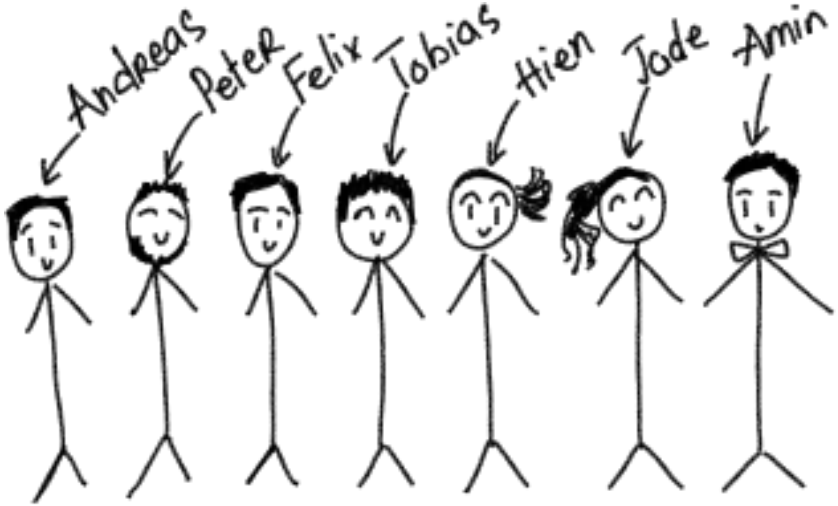

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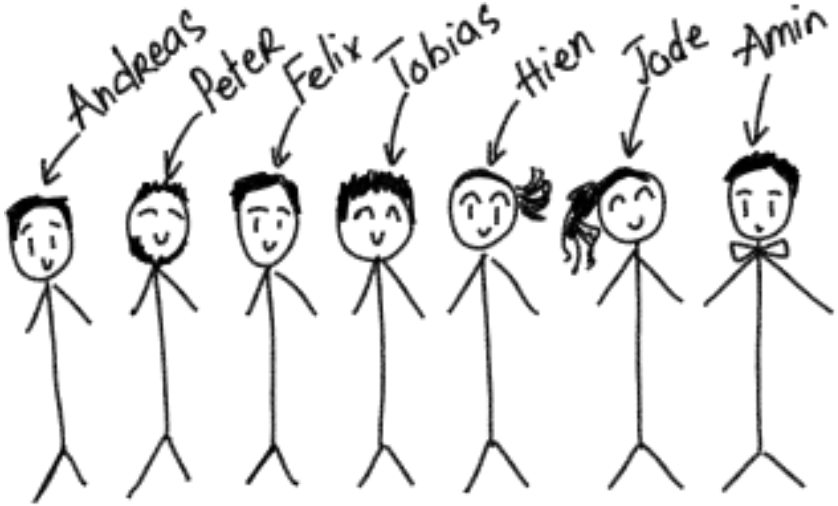
New SHA3/SHAKE hardware connection

Paper Code

Towards ML-KEM and ML-DSA on OpenTitan (IEEE S&P 2025)



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Result at target frequency

← Blog post:
Accelerating Post-Quantum Cryptography on OpenTitan-based Designs: Flexible Hardware for a Secure Future

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- **6-9x times faster** than ML-KEM and ML-DSA on OTBN without vector instructions

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- Cell-count increase of **less than 17% for OTBN** and **less than 3% for OpenTitan's Top Earlgrey (without memory)**

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But…We can still do better

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- bn.mulvm takes 12 cycles and bn.mulv takes 4 cycles

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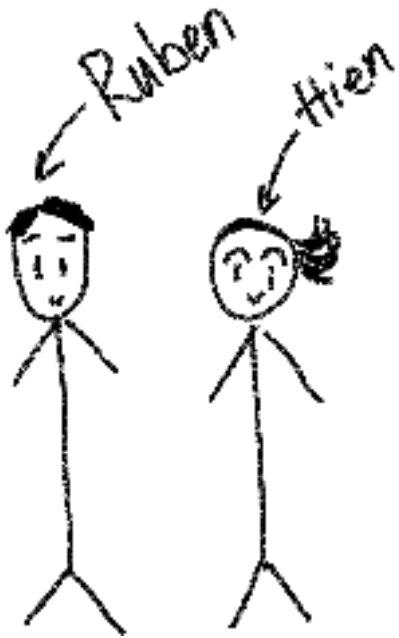
- Sequential vector adder design leads to reduced maximum frequency



Paper Code

Shift modular multiplication back to SW:

- ~~bn.mulv(m)~~ bn.mulv{.lo,.hi}{.even,.odd} -> V{1,2,3}

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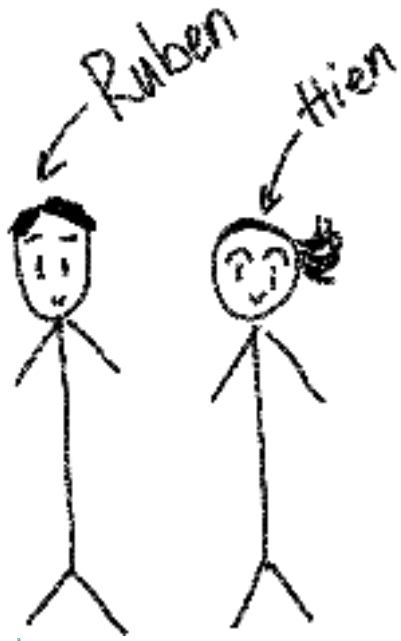

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Improved vector adder designs (tailored for target platform ASIC/FPGA)

Paper Code

Improving ML-KEM and ML-DSA on OpenTitan (CHES 2026)

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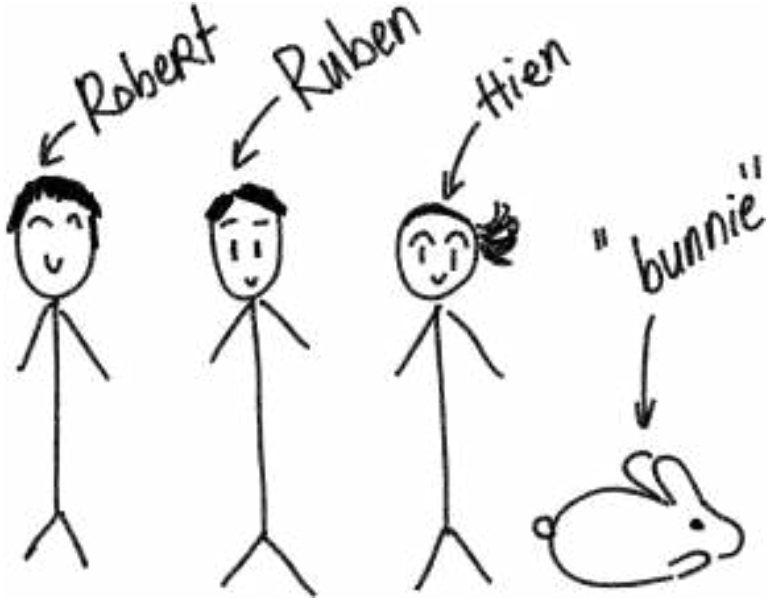

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Improved vector adder designs (tailored for target platform ASIC/FPGA)

ASIC synthesis on other technology nodes and confirmed that the results are consistent with what we had using Cadence and OpenROAD

Paper Code

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- Up to 17% speedup in cycle count and 16% increase in code size for ML-KEM and ML-DSA compared to previous paper on OTBNV2

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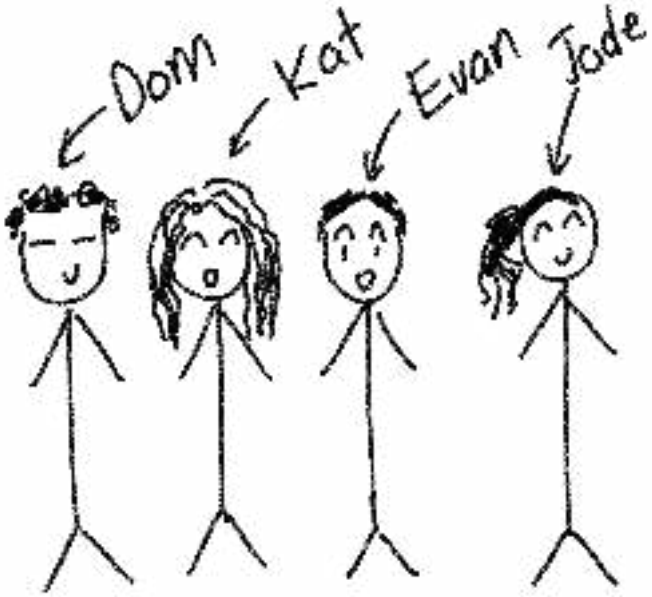
- Up to 17% speedup in cycle count and 16% increase in code size for ML-KEM and ML-DSA compared to previous paper on OTBNV2
- Similar area while improving maximum frequency of the whole OTBN

Ahem…but we can still do better!!!

Here comes the ZeroRISC refinement squad...



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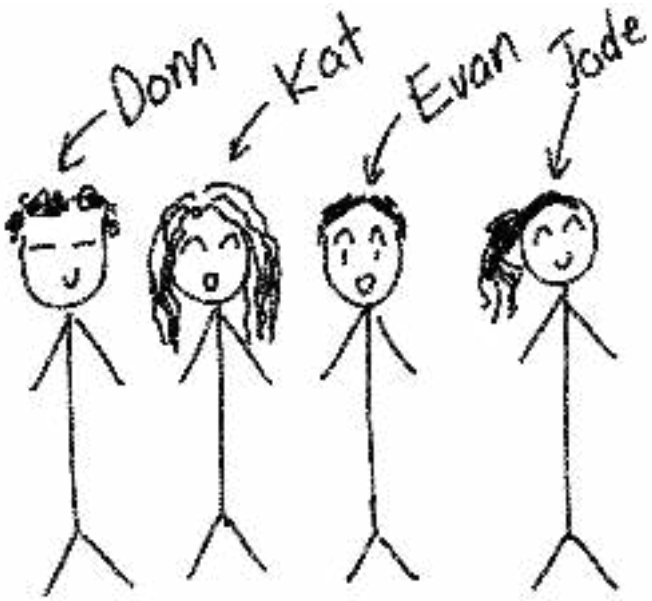

Stack optimized ML-DSA

- 121kB -> 11kB (91% reduction) for ML-DSA-87 signing
- Tradeoff: 80%-140% slowdown for sign (negligible for other ops)

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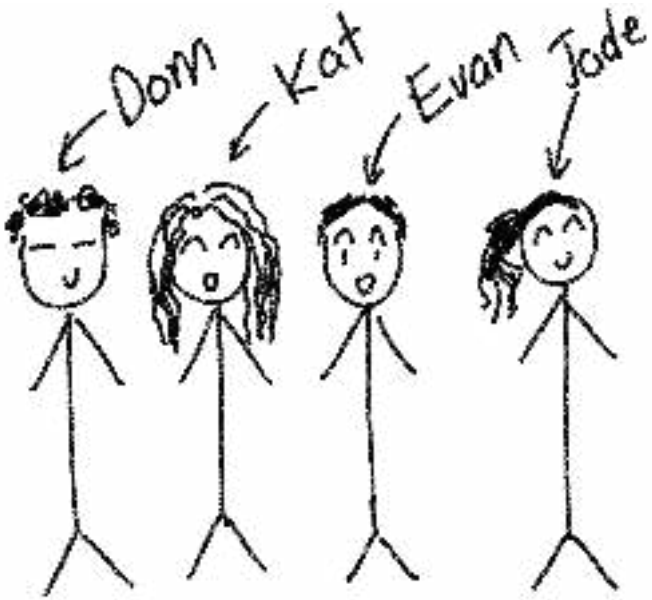

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- ML-KEM: ~13% speedup (all ops)
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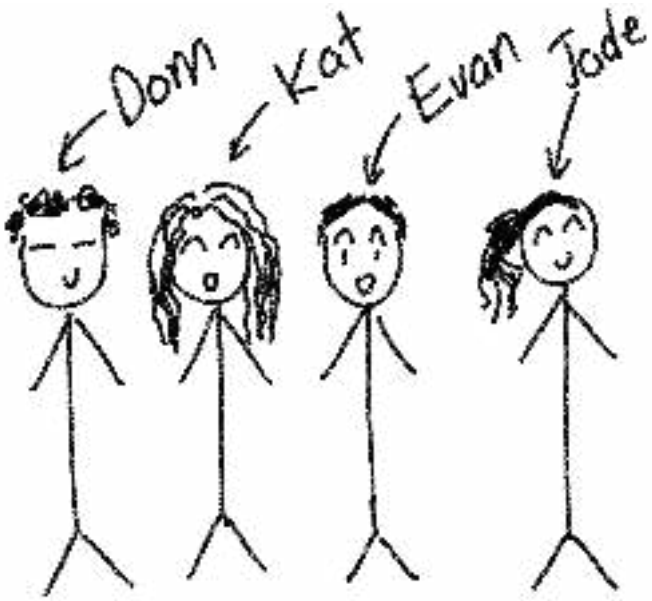

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Redesigned SHA3/SHAKE interface

- ~6% speedup (all ops)

Integration with Design Verification tools

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Beyond bignums...

OTBN + ML-DSA + ML-KEM = ACC

Asymmetric
Cryptography
Coprocessor

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OTBN + ML-DSA + ML-KEM = ACC

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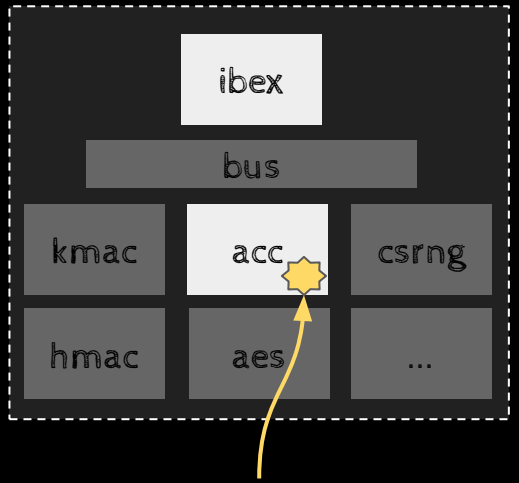
ML-KEM-768 decapsulation: 0.7ms at 100MHz
ML-DSA-65 signing (median): 6.5ms at 100MHz
Memory: 32KB IMEM, 32KB DMEM

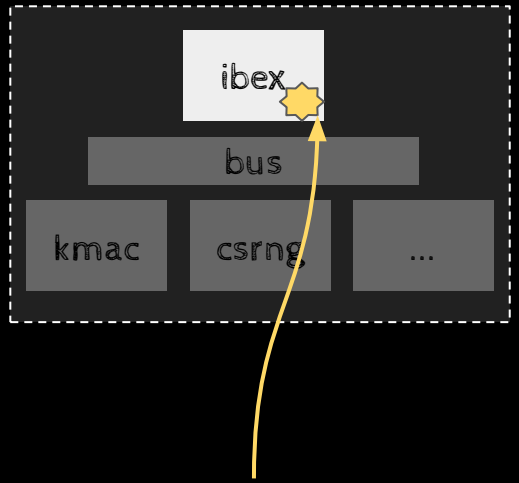
Multiple integration options for flexibility

Ibex + ACC

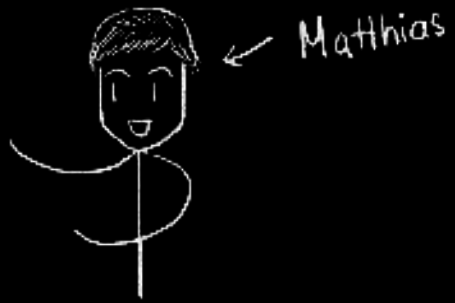


fast ML-KEM and ML-DSA on ACC

Ibex only



mlkem-native and mldsa-native on Ibex



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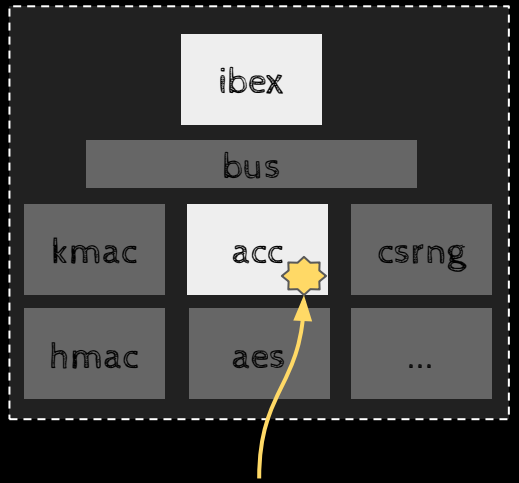
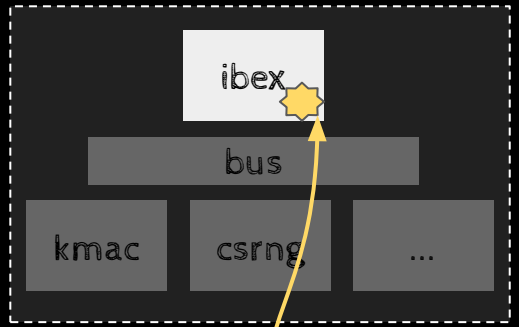


fast ML-KEM and ML-DSA on ACC

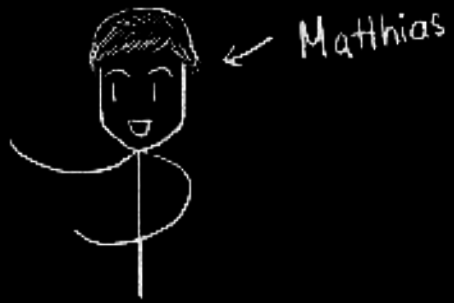
Ibex only



mlkem-native and mldsa-native on Ibex

Lower area, but slower asymmetric cryptography:

- ~10x for ML-KEM, ML-DSA
- ~20x for RSA



Ongoing Work, Lessons Learned

ACC

Preparing for
industry and
research tapeouts

Preparing for
industry and
research tapeouts

ISE for
UOV/Mayo,
FN-DSA

Preparing for industry and research tapeouts

Iasmin support for ACC

ISE for UOV/Mayo, FN-DSA

Preparing for industry and research tapeouts

Iasmin support for ACC

SLOTHY for automatic leakage mitigation

ISE for UOV/Mayo, FN-DSA

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Side-channel attacks & evaluation

Preparing for **industry and research tapeouts**

Iasmin support for ACC

SLOTHY for automatic **leakage mitigation**

ISE for **UOV/Mayo, FN-DSA**

Side-channel attacks & evaluation

Masking for ML-KEM & ML-DSA (Hien's internship)

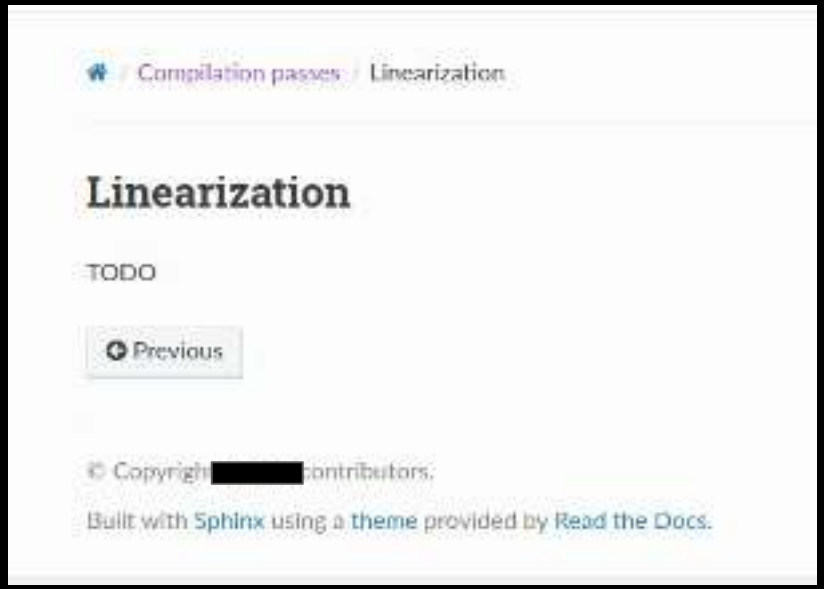
Lesson 1

Open-source removes friction

Lesson 2

Documentation saves ...lives

Official Documentation



“Official” Documentation

Hi Santiago,

Hi Santiago,

Hi Santiago,

Hellu Santiago,

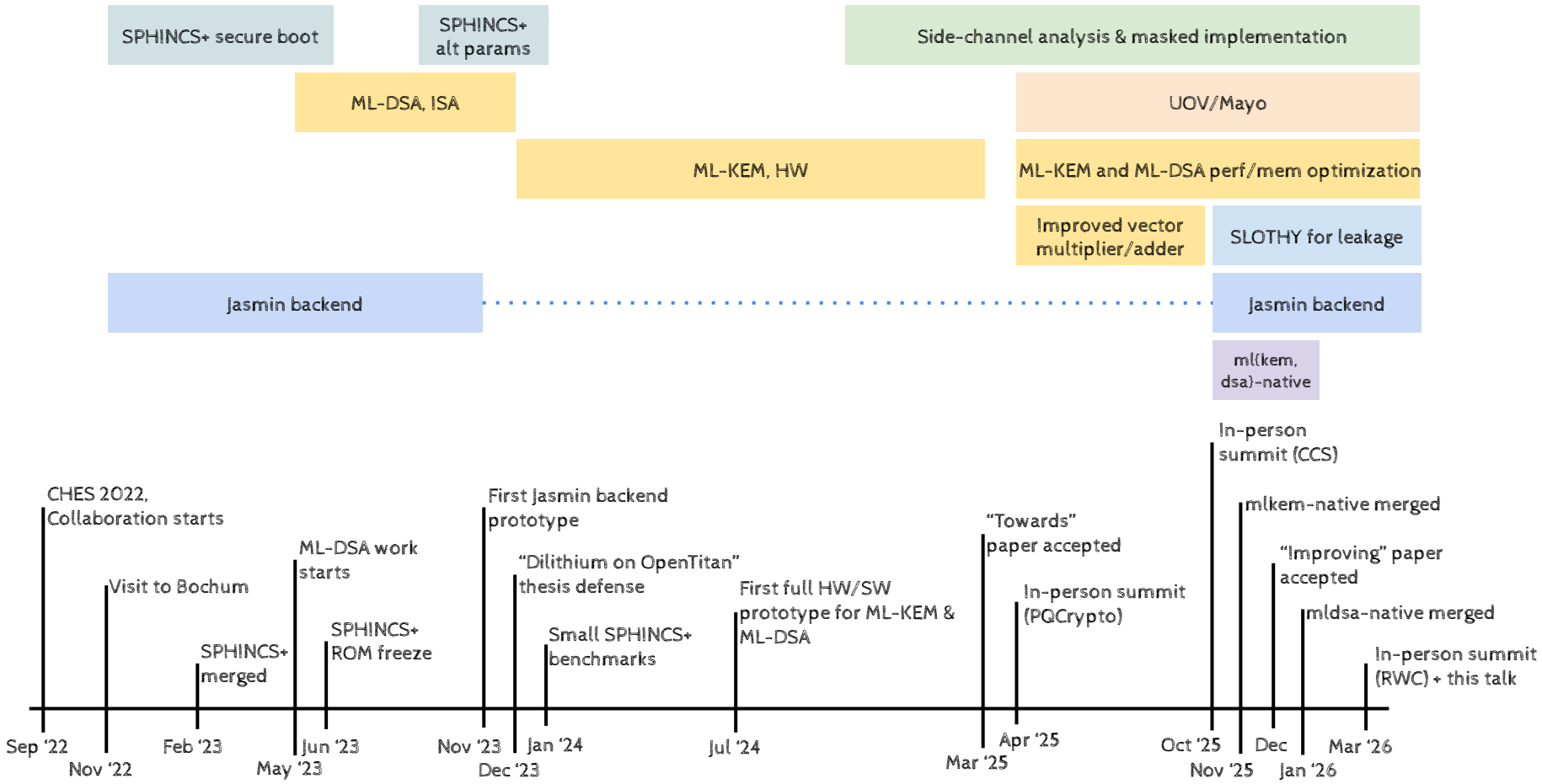
Hi Santiago (again),

Lesson 3

Research sparks, industry scales

Lesson 4

Meeting in person MATTERS!!!


SPHINCS+ secure boot
SPHINCS+ alt params
Side-channel analysis & masked implementation
ML-DSA, ISA
UOV/Mayo
ML-KEM, HW
ML-KEM and ML-DSA perf/mem optimization
Improved vector multiplier/adder
SLOTHY for leakage
Jasmin backend
Jasmin backend
ml(kem, dsa)-native
CHES 2022, Collaboration starts
Visit to Bochum
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In-person summit (CCS)
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In-person summit (RWC) + this talk
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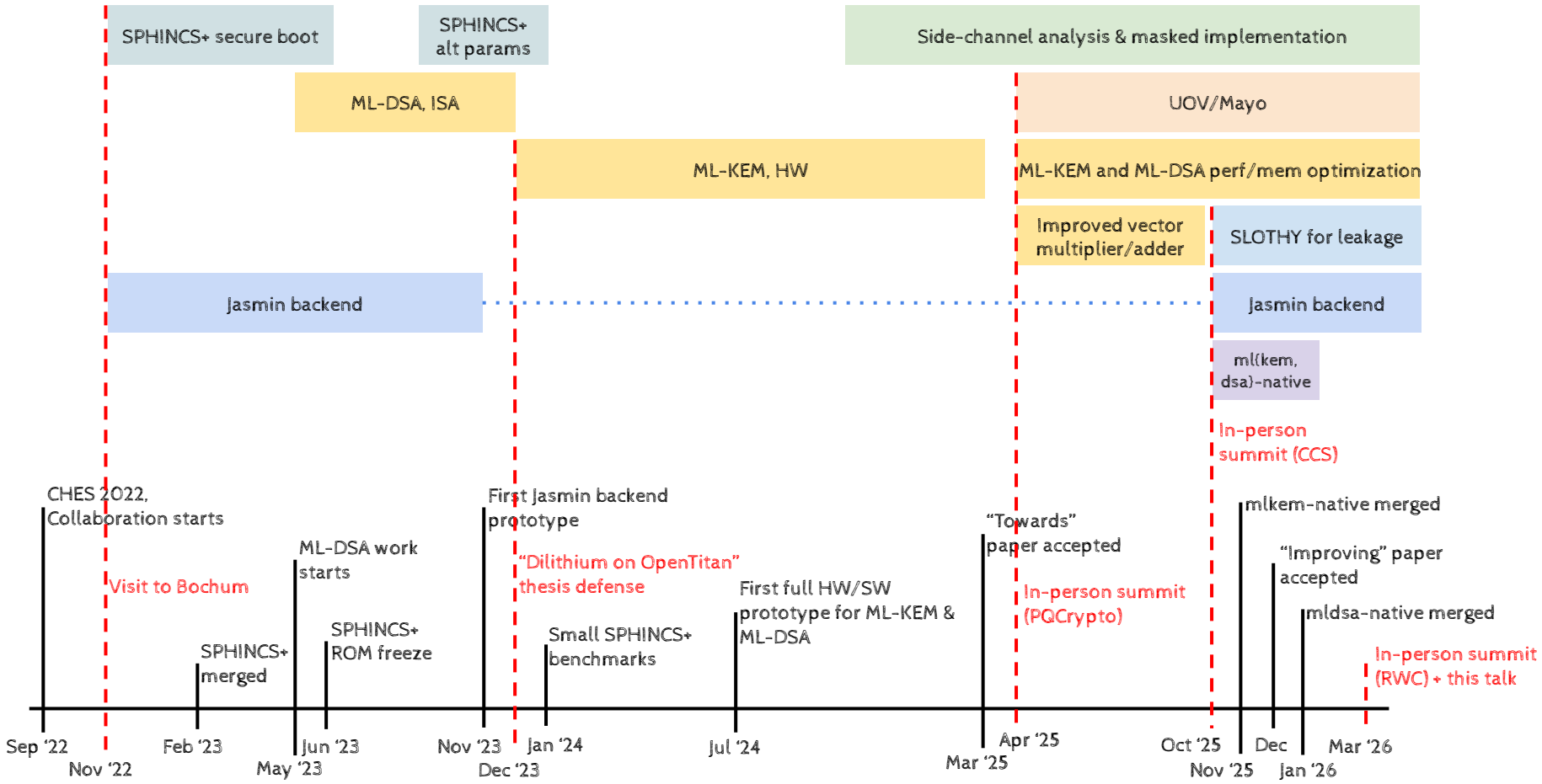
Dec
Jan '26
Mar '26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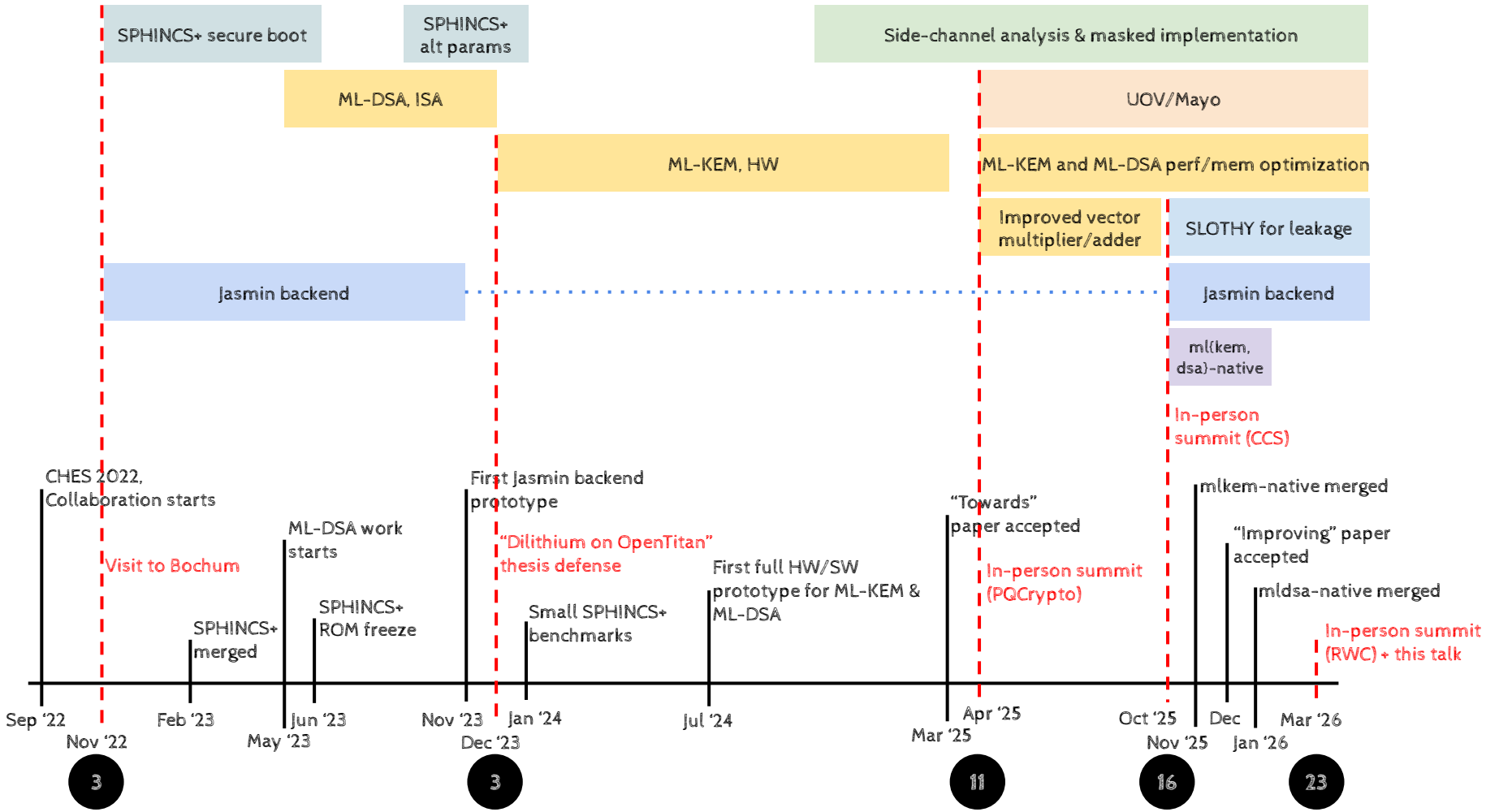
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華河街觀光夜市
Raohe St. Night Market
哈囉海鮮鍋燒

Thank you for your attention

Towards ML-KEM & ML-DSA on OpenTitan (IEEE S&P 2025)

ZeroRISC's OpenTitan fork

Improving ML-KEM & ML-DSA on OpenTitan (CHES 2026)

ZeroRISC's Blog

Max Planck Institute for Security and Privacy (MPI-SP) Blog

Presentation art: Hoang Nguyen Hien Pham

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- ZeroRISC: info@zerorisc.com